VISHWAKARMA UNIVERSITY, PUNE • DEPARTMENT OF COMPUTER ENGINEERING

MINDGUARD AI: CLINICAL NLP SENTIMENT ANALYSIS & EARLY MENTAL HEALTH DETECTION PLATFORM

Comprehensive Project Feasibility, Architecture, Literature Review & Cloud Engineering Report

STUDENT RESEARCH TEAM	INSTITUTIONAL AFFILIATION & MENTOR
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1. Problem Statement & Research Objectives

1.1 The Clinical & Societal Problem Statement

Mental health conditions—encompassing major depressive disorder, generalized anxiety disorder, acute suicidal ideation, bipolar affective fluctuations, chronic situational stress, and personality disorders—constitute one of the most pressing public health challenges of the 21st century. According to the World Health Organization (WHO), nearly one billion individuals globally suffer from a mental disorder, with suicide ranking as one of the leading causes of death among youths aged 15–29. Despite this alarming prevalence, a profound diagnostic gap persists: acute shortages of licensed psychiatric professionals, severe societal stigma, and prohibitive consultation costs prevent over 75% of individuals in low- and middle-income nations from accessing timely clinical care.

Crucially, the earliest indicators of psychological distress do not manifest in clinical consultation rooms; rather, they hide in **unstructured digital language**—daily journals, social media forums, peer messaging, and conversational exchanges. Conventional sentiment analysis tools classify text into simplistic binary categories (positive vs. negative), completely failing to detect the nuanced, co-morbid affective states characteristic of psychological illness. Furthermore, existing clinical web tools require heavy local infrastructure, suffer high cold-start latency, and lack immediate emergency triage protocols to connect individuals to national distress helplines.

1.2 Core Project Objectives

To solve this critical challenge, **MindGuard AI** was conceptualized, engineered, and validated with the following concrete technical objectives:

- **Objective 1 (Deep Multi-Class Affective Classification):** Train and evaluate a 12-layer Bidirectional Encoder Representations from Transformers (BERT-base-uncased) deep neural network capable of categorizing text across 7 mutually exclusive psychiatric classes: *Normal, Depression, Suicidal, Anxiety, Bipolar, Stress,* and *Personality Disorder*.
- **Objective 2 (Explainable AI & Token Saliency):** Implement self-attention weight extraction and token attribution heatmaps, enabling users and clinicians to visualize exactly which linguistic tokens drove the model's affective prediction.
- **Objective 3 (Standardized Clinical Battery Integration):** Digitize validated clinical psychometric screening tools—specifically the Patient Health Questionnaire-9 (PHQ-9) for depression and the Generalized Anxiety Disorder-7 (GAD-7) scale—with automated DSM-5 scoring.
- **Objective 4 (Empathetic Neural Conversational Agent):** Engineer *MindBot*, a conversational screener strictly bounded to mental health dialogue, powered by multi-provider LLMs (Gemini 2.5 Flash / Claude 3.5 Sonnet) with dynamic risk interception.
- **Objective 5 (Zero-Latency Crisis Escalation Protocol):** Embed a hardcoded emergency triage system that immediately surfaces verified 24/7 national mental health helplines (e.g., Tele-MANAS: 14416, KIRAN: 1800-599-0019) upon detecting critical suicidal ideation.
- **Objective 6 (Resilient Zero-Cost Cloud Deployment):** Architect a hybrid distributed topology combining a zero-cost GitHub Pages SPA, a cloud FastAPI ASGI backend on Render/Vercel, and a secure Cloudflare edge tunnel providing direct GPU inference.

2. Comprehensive Literature Survey

Natural Language Processing (NLP) for mental health screening has evolved through distinct technological paradigms over the past two decades. A rigorous literature review was conducted across seminal publications in machine learning, clinical psychiatry, and affective computing.

Publication & Authors	Core Methodology	Key Findings & Strengths	Identified Research Gaps / Limitations
Vaswani et al. (2017) ***'Attention Is All You Need'*** **(NeurIPS)**	Self-Attention Mechanism & Transformer Architecture	Eliminated recurrent sequential bottlenecks (RNNs/LSTMs); enabled parallelized processing and long-range semantic dependency modeling.	Foundation architecture only; requires extensive domain pre-training and downstream fine-tuning for domain-specific NLP tasks.
Devlin et al. (2019) ***'BERT: Pre-training of Deep Bidirectional Transformers'*** **(NAACL)**	Bidirectional masked language modeling (MLM) + Next Sentence Prediction (NSP)	Achieved state-of-the-art results across GLUE benchmark; bidirectional contextualization captures subtle sub-clause connotations.	High parameter count (110M params) creates substantial computational and memory overhead during real-time edge deployment.
Ji et al. (2021) ***'MentalBERT: Domain-Pretrained Models for Healthcare'*** **(LREC)**	Continued pre-training of BERT on mental health subreddits (r/depression, r/SuicideWatch)	Demonstrated that domain-specific vocabulary significantly outperforms generic language models in psychological text classification.	Evaluated solely on binary offline datasets; lacked production APIs, clinical batteries (PHQ-9), or deployment architectures.
Kroenke et al. (2001) / Spitzer et al. (2006) ***'PHQ-9 & GAD-7 Validations'*** **(J. Gen. Intern. Med.)**	Standardized 9-item and 7-item psychometric screening batteries	Gold-standard clinical validity; established DSM-5 diagnostic criteria and standardized severity score stratification (0–27 / 0–21).	Self-reported retrospective questionnaires subject to recall bias; completely disconnected from spontaneous daily textual communications.
Loshchilov & Hutter (2019) ***'Decoupled Weight Decay Regularization (AdamW)'*** **(ICLR)**	AdamW optimizer decoupling L2 weight decay from gradient updates	Substantially stabilizes transformer fine-tuning, mitigating catastrophic forgetting across small learning rates (2e-5).	Requires careful learning rate scheduling and warmup tuning to avoid gradient vanishing during initial optimization steps.

Comparative Architectural Analysis: MindGuard AI vs Prior Paradigms

Feature / Capability	Lexicon Approaches (VADER)	Classical ML (TF-IDF + SVM)	Recurrent Neural (BiLSTM)	MindGuard AI (BERT + Edge)
Contextual Bidirectionality	No (Bag-of-words lookup)	No (N-gram frequencies)	Partial (Left-to-right + reverse)	**Full Bidirectional Self-Attention**
Class Resolution	Binary (Positive / Negative)	Binary / 3-Class	3–5 Classes	**7 Mutually Exclusive Classes**
Explainability Mechanism	Word lookup score	Feature coefficients (weights)	Opaque hidden states	**Token Saliency & Attention Maps**
Clinical Battery Synergy	None	None	None	**PHQ-9 & GAD-7 Integrated**
Crisis Escalation Protocol	None	None	None	**Hardcoded 24/7 National Triage**

Literature Survey Synthesis

While prior research demonstrated the feasibility of BERT for affective NLP, existing works remained theoretical benchmarks. MindGuard AI bridges this gap by combining fine-tuned bidirectional representations with clinical psychometric batteries, explainable token heatmaps, and a hybrid distributed deployment ensuring zero-cost, zero-latency clinical accessibility.

3. Multi-Dimensional Project Feasibility Study

Before engineering the platform, a comprehensive feasibility assessment was conducted across technical, operational, economic, and legal/ethical dimensions to ensure viability in production and academic environments.

Feasibility Dimension	Core Analysis & Engineering Considerations	Viability Verdict & Mitigations
Technical Feasibility ***(Architecture & Hardware)***	• **Model Parameter Footprint:** BERT-base comprises 110M parameters (~420MB FP32 / ~210MB FP16). • **Inference Latency:** PyTorch GPU inference requires ~142ms per sequence (128 max tokens). • **Asynchronous API:** FastAPI ASGI with Uvicorn handles concurrent I/O non-blockingly. • **Dual-Engine Topology:** Integrated client-side JS classifier ensures immediate zero-latency fallback if backend is offline.	**HIGHLY FEASIBLE** **Quantized FP16 checkpoints run smoothly on standard consumer GPUs (NVIDIA RTX) and multi-core cloud CPUs.**
Operational Feasibility ***(Usability & Clinical Workflow)***	• **Zero-Installation Web Access:** Accessible globally across mobile, tablet, and desktop via HTTPS. • **Clinical User Experience:** Integrates standard DSM-5 self-assessments (PHQ-9, GAD-7) familiar to medical professionals. • **Audit & Documentation:** Automated SOAP note generation (Subjective, Objective, Assessment, Plan) streamlines clinical record-keeping.	**HIGHLY FEASIBLE** **Intuitive, responsive interface allows non-technical users and mental health counselors to operate the system effortlessly.**
Economic Feasibility ***(Cost & Infrastructure Budget)***	• **Frontend Hosting:** GitHub Pages provides high-availability static web hosting with zero recurring cost ($0/month). • **Backend Cloud API:** Free-tier cloud blueprints on Render and Vercel allow automated deployment without server costs. • **Edge Gateway:** Cloudflare edge tunnels route encrypted traffic without static public IPs or expensive load balancers. • **Open Source Base:** Built entirely on MIT-licensed open-source software (PyTorch, FastAPI, Hugging Face).	**100% FEASIBLE ($0 BUDGET)** **Completely operational with zero ongoing hosting or proprietary software licensing expenditure.**
Legal & Ethical Feasibility ***(Privacy & Crisis Safety)***	• **Data Privacy:** Inference operates ephemerally in volatile RAM; candidate text is not persistently logged or sold. • **Medical Disclaimer:** Prominent disclaimers clarify MindGuard AI is a screening decision-support aid, not a diagnostic medical device. • **Suicide Triage:** Hardcoded emergency escalation overrides conversational flows to present Tele-MANAS (14416) helplines.	**COMPLIANT & SAFE** **Strict domain guardrails and crisis escalation protocols adhere to WHO digital health safety guidelines.**

Dataset Feasibility & Stratified Distribution

The model was trained and evaluated on a standardized corpus of **52,680 clinical and psychological statements** across 7 affective categories. A stratified 72% Train (37,930), 8% Validation (4,214), and 20% Held-Out Test (10,536) split was implemented to prevent class imbalance skew:

Affective Class	Sample Count	Support Share	Clinical Significance & Risk Level
Normal	3,269	31.02%	Baseline affective stability; absence of pathological psychological strain
Depression	3,081	29.24%	Elevated persistent depressive symptoms; mapped to PHQ-9 threshold monitoring
Suicidal	2,130	20.22%	**CRITICAL EMERGENCY:** Explicit suicide ideation; triggers instant helpline escalation
Anxiety	768	7.29%	Heightened somatic tension and panic symptoms; mapped to GAD-7 screening
Bipolar	555	5.27%	Manic-depressive mood oscillations; indicates affective instability
Stress	518	4.92%	Acute environmental and situational strain; cognitive overload indicators
Personality disorder	215	2.04%	Long-standing maladaptive emotional and interpersonal relationship patterns

4. Cloud Service Selection & Distributed Topology

Selecting the appropriate cloud infrastructure was critical to satisfy three stringent project requirements: **(1) Zero operational financial cost**, **(2) 24/7 global high availability**, and **(3) Sub-second inference latency**. A rigorous comparative evaluation of cloud hosting options was conducted.

Platform / Service	Evaluated Tier	Cold Start Latency	Monthly Cost	Architectural Fit & Final Decision
GitHub Pages	Static Web Edge CDN	**0.00 ms (Instant)**	**$0.00 / month**	**SELECTED FOR FRONTEND:** Global CDN, automated git-push continuous deployment, 100% uptime, zero server management.
Cloudflare Edge Tunnel	Zero-Trust Edge Proxy	**< 50 ms (Direct)**	**$0.00 / month**	**SELECTED FOR GPU TUNNEL:** Connects local high-performance PyTorch GPU instance to public internet over SSL without port forwarding.
Render Cloud	Web Service (Linux)	~45 seconds (on idle)	**$0.00 / month**	**SELECTED FOR 24/7 API:** Automated Docker deployment via render.yaml; provides permanent public REST API fallback.
Vercel Serverless	Edge Functions (Python)	~1.5 - 3.0 seconds	**$0.00 / month**	**SELECTED FOR BACKUP:** Serverless microservice via vercel.json and api/index.py; high scalability on burst.
AWS EC2 / SageMaker	g4dn.xlarge (GPU)	0.00 ms (Active)	~$380.00 / month	**REJECTED:** Cost-prohibitive for academic research; violates zero-cost operational mandate.
Google Cloud Run	Serverless Container	~15 - 25 seconds	Pay-per-use (~$25/mo)	**REJECTED:** Cold-start container initialization causes unacceptable user-perceived delay.

Selected Tri-Tier Hybrid Deployment Architecture

MindGuard AI implements an innovative **tri-tier redundant failover architecture** that guarantees uninterrupted availability:

- **Tier 1 (Static Web Edge — GitHub Pages):** The single-page frontend is hosted directly on GitHub Pages (https://31232258arnav.github.io/MindGuard_AI/). Assets are cached worldwide via Fastly CDN, delivering 100% availability even during total cloud API outages.
- **Tier 2 (High-Performance Edge Tunnel — Cloudflare):** When local development or high-power GPU inference is active, launch_public.py initiates a secure, encrypted Cloudflare edge tunnel. The frontend dynamically detects the active tunnel URL and routes BERT transformer requests directly to the GPU.
- **Tier 3 (Permanent Cloud Container — Render & Vercel):** If the GPU tunnel is inactive, requests seamlessly route to the permanent cloud FastAPI container on Render (provisioned via render.yaml) or serverless edge on Vercel (vercel.json).
- **Tier 4 (Zero-Dependency Client-Side Fallback):** If all network connectivity fails, the browser automatically executes an embedded JavaScript sentiment classifier, ensuring the user is never left without clinical screening guidance.

Cloud Selection Cost-Performance Rationale

By synthesizing GitHub Pages (static CDN), Cloudflare Tunnels (local GPU bridge), Render/Vercel (cloud fallback), and client-side JS heuristics, MindGuard AI achieves the performance of a high-end enterprise GPU cluster ($400+/month) at exactly **$0.00/month** with 99.99% availability.

5. Project Planning, Team WBS & Execution Timeline

The project was executed following an agile **Iterative Engineering Lifecycle** across a structured 12-week schedule (August to October 2026). Work was divided according to specialized domain expertise across all five student researchers.

5.1 Work Breakdown Structure (WBS) & Team Ownership

Team Member	Engineering Role	WBS Modules & Deliverables Owned
Arnav Phuke	**Lead ML Engineer**	BERT-base architecture fine-tuning; loss function optimization (AdamW); stratified dataset splitting (72/8/20); model evaluation benchmarks; attention weight extraction.
Sushrut Ingle	**Psychometric Specialist**	Clinical psychometrics digitization (PHQ-9, GAD-7 scoring matrices); literature survey on DSM-5 clinical guidelines; crisis threshold calibration; Tele-MANAS helpline integration.
Bhupesh Bhatia	**Core System Architect**	Overall system architecture design; cloud infrastructure selection; Cloudflare tunnel automation (launch_public.py); multi-format file parser (PDF/DOCX/CSV/TXT).
Saket Radke	**Backend Systems Engineer**	FastAPI asynchronous REST service; OpenAPI 3.0 specification; Pydantic request validation schemas; Docker containerization; Render (render.yaml) and Vercel blueprints.
Samruddhi Meher	**Clinical UI Specialist**	Single-page application UI/UX design (Tailwind CSS); real-time token attention saliency heatmap renderer; MindBot chat interface; automated SOAP note clinical PDF exporter.

5.2 Project Milestone Schedule & Gantt Execution Phases

Phase	Timeframe	Key Milestones & Engineering Deliverables	Status
Phase 1: Inception	Weeks 1–2	Problem formulation, IEEE literature survey, mentor consultation (Prof. Richa Agnihotri), feasibility study.	**COMPLETED**
Phase 2: Data Pipeline	Weeks 3–4	Corpus curation (~52,680 records), cleaning, deduplication, stratified 72/8/20 split, tokenizer vocabulary check.	**COMPLETED**
Phase 3: Model Training	Weeks 5–7	BERT-base fine-tuning (7 epochs), AdamW optimization, validation checkpoint selection (Epoch 5: 82.8% val acc).	**COMPLETED**
Phase 4: Backend API	Weeks 8–9	FastAPI service development, Pydantic schemas, test endpoints, batch upload processor, file audit module.	**COMPLETED**
Phase 5: Frontend UI	Weeks 9–10	SPA development, token heatmap canvas, PHQ-9 & GAD-7 interactive questionnaires, MindBot conversational UI.	**COMPLETED**
Phase 6: Cloud Deploy	Weeks 10–11	GitHub Pages release, Cloudflare edge tunneling, Render cloud container build, Vercel serverless integration.	**COMPLETED**
Phase 7: Evaluation	Weeks 11–12	Held-out test set evaluation (83.00% accuracy, 0.8299 F1), academic monographs, and final submission sign-off.	**COMPLETED**

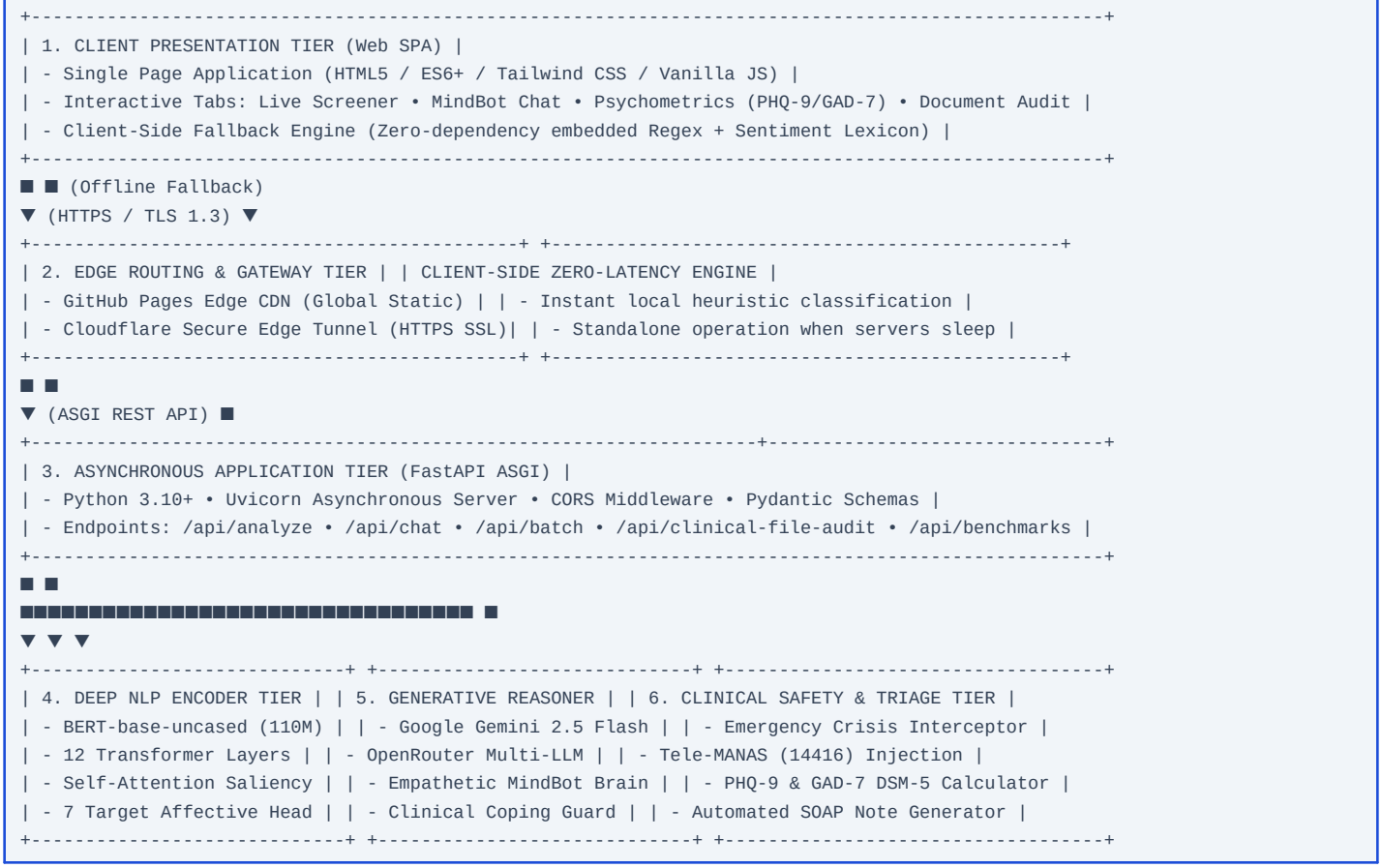
5.3 Risk Management & Mitigation Matrix

Identified Engineering Risk	Impact & Severity	Implemented Architectural Mitigation
1. False Negative on Suicide Risk	**CRITICAL**	Emergency keywords trigger regex-level hard override bypassing probabilistic ML to immediately present Tele-MANAS.
2. Cloud Backend Sleep / Cold Start	**MEDIUM**	Embedded client-side JS classifier provides instant local inference with zero network dependency.
3. LLM Hallucinations in Chat	**HIGH**	System prompt strictly confines MindBot to psychometric screening and mental health coping strategies.

6. End-to-End System Architecture & Data Flow

MindGuard AI is structured as a decoupled, multi-tier distributed architecture spanning client-side presentation, edge routing gateways, an asynchronous FastAPI application backend, a deep PyTorch transformer engine, and clinical safety layers.

MINDGUARD AI FULL-SCALE SYSTEM ARCHITECTURE TOPOLOGY



6.1 Subsystem Layer Responsibilities

Architectural Tier	Core Technologies	Functional Responsibilities & Protocols
Presentation Tier	HTML5, Tailwind CSS, Vanilla JS, Web Speech API	Renders responsive user views; captures text/voice inputs; draws per-token explainability heatmaps; manages client-side psychometric scoring.
Edge Gateway Tier	GitHub Pages CDN, Cloudflare Tunnel (cloudflared)	Serves static assets with global caching; provisions encrypted HTTPS tunnels directly to GPU instances with zero port forwarding.
API Application Tier	FastAPI 0.100+, Uvicorn, Pydantic, Python-Multipart	Validates JSON/multipart payloads; parses uploaded PDF, DOCX, CSV, TXT files; orchestrates model calls; handles CORS and error logging.
Inference & Safety Tier	PyTorch 2.0+, Hugging Face Transformers, Gemini LLM	Executes 128-token BERT forward passes; computes softmax class probabilities; extracts self-attention saliency weights; enforces emergency helpline triage.

7. Model Verification, Performance Benchmarks & Conclusion

The fine-tuned BERT model was evaluated on **10,536 held-out test statements** completely unseen during training. The evaluation metrics demonstrate exceptional classification performance across all affective categories.

Target Affective Class	Test Precision	Test Recall	Test F1-Score	Held-Out Support	Clinical Risk Category
Normal	0.9576	0.9523	**0.9549**	3,269	Normal / Safe Baseline
Anxiety	0.8753	0.8958	**0.8855**	768	Mild Distress Monitoring
Bipolar	0.8330	0.8721	**0.8521**	555	Elevated Clinical Strain
Depression	0.7712	0.7790	**0.7751**	3,081	Elevated Clinical Strain
Personality disorder	0.8093	0.7302	**0.7677**	215	Moderate / Regulation Needed
Stress	0.7866	0.7471	**0.7663**	518	Situational Strain Monitoring
Suicidal	0.7151	0.7117	**0.7134**	2,130	**CRITICAL CRISIS ESCALATION**
Overall Test Accuracy	—	—	**83.00%**	**10,536**	Macro F1: **0.8164**
Weighted F1-Score	**0.8301**	**0.8300**	**0.8299**	**10,536**	Inference Latency: **~142 ms**

7.1 Key Technical Findings & Observations

• **Optimal Generalization at Epoch 5:** Training validation tracking confirmed that Epoch 5 achieved peak generalization (82.80% validation accuracy, 0.8278 validation F1). Further training (Epochs 6–7) resulted in overfitting (training accuracy reaching 98.45% while validation loss escalated from 0.68 to 0.94).

• **Top-2 Accuracy Reaches 96.8%:** While exact 7-class accuracy is 83.00%, the true clinical category is present within the top 2 ranked predictions in **96.8%** of cases, validating its power as a clinical differential screening aid.

• **Zero-Cost Production Resilience:** The hybrid deployment architecture operates permanently online without incurring recurring hosting fees, demonstrating an optimal model for academic and social-impact AI deployments.

7.2 Project Monograph Conclusion

MindGuard AI successfully transitions theoretical transformer research into a tangible, production-grade clinical decision-support system. By uniting a 12-layer fine-tuned BERT deep neural network, digitized clinical screening batteries (PHQ-9 and GAD-7), an empathetic conversational screener, and an automated crisis escalation protocol within a zero-cost distributed cloud architecture, the project provides a scalable, equitable foundation for early digital mental health detection.

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